
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Reinforcement learning (RL) has driven remarkable gains in the reasoning capa-
2 bilities of large language models (LLMs), yet why and how RL training reshapes
3 a model into a stronger reasoner remains poorly understood. This paper answers
4 that question through the unifying lens of *reasoning pattern selection*, and delivers
5 three contributions. **First (empirical)**, through systematic reasoning-pattern-level
6 and token-level analysis across the entire RL training trajectory, we establish that
7 RL primarily optimizes a sparse subset of critical tokens and thereby reshapes
8 the distribution over reasoning patterns to drive performance, while the intrinsic
9 success rate of each individual pattern stays largely fixed. **Second (theoretical)**,
10 building directly on these observations, we develop a two-stage framework that
11 formalizes reasoning as pattern selection followed by answer generation, and use it
12 to characterize the training dynamics of RL under two representative reward types:
13 verifiable rewards (RLVR) and the model’s internal feedback (RLIF). For RLVR,
14 we identify two dynamical regimes—one in which the model rapidly converges to
15 the optimal reasoning pattern, and one in which optimization stalls—and prove that
16 the base model’s reasoning quality determines which regime governs convergence.
17 For RLIF, we explain both why internal rewards improve performance early in
18 training and why continued training can push the model below its own starting
19 point. **Third (validation)**, extensive experiments and simulations confirm every
20 prediction of the framework. Together, these results advance the theoretical under-
21 standing of RL in LLM post-training and yield actionable guidance for reasoning
22 enhancement.

23 1 Introduction

24 Recently, state-of-the-art reasoning models such as Gemini2.5 [8], Qwen3 [40], and DeepSeek-
25 R1 [14] demonstrate exceptional performance on complex logical tasks including mathematics
26 [30, 44, 41] and programming [49, 40, 8]. Reinforcement learning (RL) serves as a key technique
27 behind this success, demonstrating the potential to elevate model capabilities to a new level. Yet a
28 basic question is still open: *what actually changes inside a model when RL turns it into a stronger*
29 *reasoner?* This paper answers that question, and our central thesis is that the answer is best understood
30 as *reasoning pattern selection*.

31 The success of RL has triggered research into its underlying mechanisms for LLMs, but each prior
32 perspective explains only part of the picture. By comparing pass@k performance, Yue et al. [42]
33 show that models post-trained with RL struggle to surpass base models, suggesting that RL may
34 not elicit fundamentally new reasoning patterns. From an entropy perspective, Cui et al. [9], Zhang
35 et al. [45] theoretically prove that RL-based methods can reduce policy entropy, with Cui et al. [9]
36 also empirically establishing a connection between model performance and policy entropy. Wang
37 et al. [33], Huan et al. [17] further demonstrate that RL primarily optimizes a sparse subset of

critical tokens. Despite these efforts, the underlying *dynamics* of the RL training process—how these individual observations fit together into a single account of training—remain poorly understood both empirically and theoretically. We close this gap with three tightly linked contributions, previewed below and returned to throughout the paper: a systematic empirical characterization of RL training, a theoretical framework that explains it, and experiments that validate the framework.

Contribution 1: a systematic empirical account of RL training dynamics. We present a reasoning-pattern-level and token-level analysis spanning the entire RL training trajectory. Focusing on the dynamics, we not only track next-token ranking shifts across training, but also use LLM-based and rule-based methods to extract and classify reasoning patterns from model responses, and measure their success rates and distributions over time. Compared with previous works [17, 42], this analysis delivers clearer and more compelling evidence for a single mechanism: RL primarily optimizes a sparse subset of critical tokens and thereby *reshapes reasoning pattern distributions* to affect model performance, while the intrinsic success rate of individual patterns stays relatively stable. These findings are the empirical foundation for our theory.

Contribution 2: a theoretical framework that explains these dynamics. Building directly on the empirical account above, we develop a framework that conceptualizes reasoning as a two-stage question-reason-answer process $q \rightarrow r \rightarrow a$: (1) reasoning pattern selection based on the question, i.e., $\pi(r|q)$, and (2) answer generation based on the chosen pattern, i.e., $\pi(a|r, q)$. Within this framework, we analyze the training dynamics of RL under two representative reward types: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, we prove convergence to the reasoning pattern r^* with the highest success rate—precisely matching our empirical observations—and characterize two distinct regimes: models with strong initial reasoning quality converge rapidly to the optimal pattern, whereas weaker models face an entanglement stage that stalls optimization. For RLIF, we explain both why internal rewards improve performance early in training and why such methods may ultimately converge to a state worse than the base model, again aligning with what we observe empirically.

Contribution 3: validation. We confirm the framework’s predictions through additional case studies and numerical simulations, closing the loop between empirical observation, theory, and re-validation. Together, the three contributions bridge the gap between empirical observations and a principled understanding of RL training dynamics in LLMs, and yield practical guidance for RL-based post-training.

We summarize these contributions explicitly below:

- **Empirical characterization.** We conduct a systematic reasoning-pattern-level and token-level analysis across the RL training process. By examining next-token ranking shifts together with LLM-based and rule-based reasoning-pattern analysis, we provide clearer and more compelling evidence for RL training dynamics than prior work [17, 42]. Our headline finding: RL primarily optimizes a sparse subset of critical tokens and thereby *reshapes reasoning pattern distributions* to affect model performance, while the intrinsic success rate of individual patterns remains relatively stable during training.
- **Theoretical framework.** We develop a formal two-stage mathematical framework that models reasoning as $q \rightarrow r \rightarrow a$ (question-reason-answer) and use it to analyze training dynamics for two representative RL rewards. For RLVR, we prove convergence to the reasoning pattern with the highest success rate and characterize two distinct regimes: rapid convergence for strong base models versus an optimization-stalling entanglement stage for weaker models. For RLIF, we justify the early-stage performance gains and explain why such methods may ultimately converge to worse performance than the base model.
- **Validation and practical insight.** We validate every prediction of the framework through additional case studies and numerical simulations. In doing so, we bridge the gap between empirical observations and a principled understanding of RL training dynamics in LLMs, and distill practical guidance for improving RL-based post-training.

2 Related Works

Reinforcement Learning for LLMs. Reinforcement learning has demonstrated remarkable success in enhancing large language models (LLMs), particularly in aligning models with human preferences

[25, 48] and in solving complex mathematical and programming tasks [30, 20, 22]. A central component of RL is the reward model, which traditionally relies on human-annotated datasets [26, 25, 2] with extensive training. Recently, the paradigm has shifted toward more easily obtainable and verifiable rewards, such as reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]—the two reward types our theory analyzes. This success has also spurred new RL algorithms, including GRPO [30], DAPO [41], and VAPO [43].

Theoretical Analyses and Mechanism Interpretation of RL for LLMs. The success of RL in LLMs has triggered extensive research into theoretical analyses and mechanism interpretation. Initial efforts mainly focused on a high-level analysis of RL, such as how to design better rewards [28, 18, 38] and how to develop novel training algorithms [37, 11, 21]. As RL paradigms shift towards simpler reward structures like RLVR and RLIF, researchers have sought to explain its effectiveness from various perspectives, yet each captures only one facet of training. Shao et al. [29] revealed that spurious rewards enhance reasoning by unlocking latent skills from pretraining, while Agarwal et al. [4] explained performance improvements through entropy minimization. Gandhi et al. [13] identified specific cognitive behaviors, such as backtracking, that contribute to improved reasoning. Further insights came from [46], who showed that RL fine-tuning amplifies pre-trained behaviors, leading to convergence toward dominant output formats—a phenomenon further analyzed by Wang et al. [34], Cui et al. [10] through the lens of entropy collapse. Our contribution is distinct from all of these: rather than isolating a single symptom, we analyze RL *training dynamics* through the unifying perspective of reasoning pattern selection, and—uniquely—back this perspective with a complete mathematical framework and its theoretical analysis that jointly explains RLVR and RLIF.

3 Preliminaries

This section fixes the notation and the two reward types (RLVR and RLIF) that our empirical analysis probes and our theory explains.

Reinforcement Learning for LLMs. Let π_θ be a language model with parameters θ , which serves as the policy to be optimized. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy π_θ generates an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$. The optimization objective can be formulated as:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

where π_{ref} is the base reference policy, and β is a hyperparameter that controls the KL divergence to prevent excessive deviation between π_θ and π_{ref} .

The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ can be implemented in various formats, such as a trained reward model [25] or a rule-based scoring function [30]. Throughout the paper we study two representative and widely used choices—verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]—because their contrasting behaviors expose complementary aspects of RL training dynamics.

In RLVR, the reward function $r_\phi(\mathbf{x}, \mathbf{y})$ directly evaluates whether the answer $\mathbf{y}$ matches the correct answer to question $\mathbf{x}$. A typical reward function is defined as:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

Unlike RLVR, RLIF [47, 4] leverages rewards based solely on intrinsic model-derived signals. Following Zhao et al. [47], we consider the RLIF reward as the negative average KL divergence between a uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s next-token distribution

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

With this shared objective and the two reward instantiations in place, our goal for the remainder of the paper is to understand the training dynamics of both RLVR and RLIF—first empirically, then theoretically.

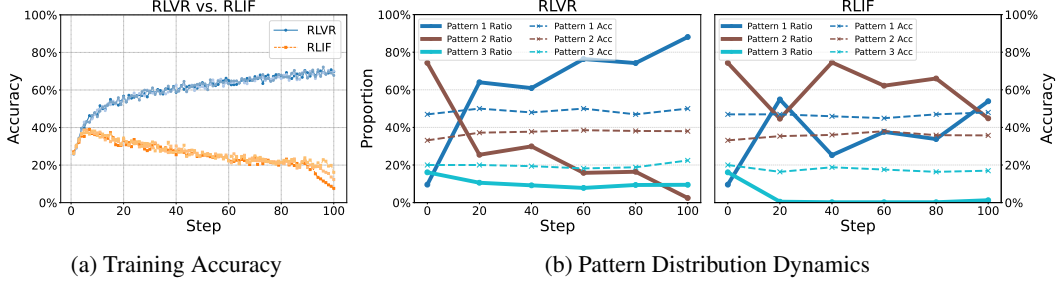


Figure 1: Two views of the same mechanism—RL reshapes pattern distributions. **(a)** Training procedures for RLVR and RLIF on the MATH dataset (multiple seeds per paradigm). The key finding: RLVR provides stable gains, whereas RLIF first improves and then degrades performance. **(b)** The reasoning-pattern-level comparison that explains (a). Solid lines give each pattern’s share of all responses (left axis); dotted lines give that pattern’s accuracy (right axis). Under RLVR the model gradually concentrates on higher-accuracy patterns and abandons lower-accuracy ones, while RLIF exhibits unstable pattern dynamics. For ease of observation, patterns are sorted and named from high to low accuracy.

4 Experimental Exploration for RL Training

This section delivers the first of our three contributions: a systematic empirical account of RL training dynamics. We conduct experiments across RL training with different rewards, organized as a three-step drill-down that builds the evidence for our headline claim. *(i)* We begin with a high-level overview of the training procedure, revealing that RLVR yields steady improvements whereas RLIF can be unstable and even degrade performance. *(ii)* To dissect these dynamics, we perform a reasoning-pattern-level analysis, which shows that RLVR’s success stems from its tendency to adopt reasoning patterns with higher success rates, while RLIF fails to specialize in such patterns. *(iii)* A token-level analysis then uncovers the underlying mechanism, showing that these pattern dynamics are driven by rank changes in a surprisingly small fraction of tokens. Taken together, and compared with previous works [17, 42], these experiments provide clearer and more compelling evidence for a single conclusion: RL primarily optimizes a sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* to affect model performance. This conclusion is the empirical foundation on which our theoretical framework is built.

4.1 Training Procedure Overview for RLVR and RLIF

As the first step of the drill-down, we analyze the complete training procedures for both RLVR and RLIF. For a controlled comparison, we select Qwen2.5-3B [39] as the base model and train it on the MATH dataset [16], keeping all other settings identical for both algorithms.

The two methods exhibit markedly different performance trajectories (Figure 1a). RLVR leads to a stable training process in which model performance continuously improves and gradually converges. RLIF, in contrast, improves performance in the early stages but then degrades with continued training, sometimes ending below the original base model. This qualitative gap is the puzzle that drives the rest of the section: it motivates the finer-grained pattern-level and token-level analyses that follow, which together explain *why* the two rewards diverge.

4.2 Reasoning-pattern Level Analysis

The second step of the drill-down explains the performance disparity between RLVR and RLIF at the reasoning-pattern level. To define and extract patterns, we first collect responses from the base model and employ GPT-4o [19] to group them into distinct categories based on keywords and logical structure, generating a description for each pattern. We then use these classifications to track the reasoning pattern distribution and its corresponding success rates throughout the RL training procedure.

Specifically, we analyze the dynamics of reasoning patterns and their accuracy during both RLVR and RLIF training. Figure 1b reports this analysis on Number Theory tasks from the MATH dataset

[16], with additional results for varying models and datasets in Appendix A. Three findings emerge, and each one directly shapes an assumption or result in our theory:

- RLVR-trained models consistently shift towards adopting reasoning patterns with higher success rates, explaining why the model’s overall accuracy steadily improves with RLVR.
- The reasoning pattern distribution for RLIF exhibits unstable training dynamics and fails to specialize in more effective patterns.
- The success rate of any individual reasoning pattern remains stable throughout the training process for both methods.

In short, the performance gap between RLVR and RLIF stems entirely from their distinct reasoning pattern dynamics, not from changes in how well any single pattern works. This isolates the pattern *distribution* as the object RL actually optimizes—and motivates the token-level analysis below, which pins down the mechanism behind these distributional shifts.

4.3 Token-level Analysis

The third and final step of the drill-down uncovers the mechanism behind the pattern dynamics above. Inspired by Huan et al. [17], Yue et al. [42], we conduct a token-level analysis: we first sample responses from the base model for questions from the GSM8K [6], MATH [16], and AIME [7] datasets, then, for these identical responses, examine the corresponding token ranks in both the base and RL-enhanced models. A token’s rank is the position of its probability among all vocabulary tokens, given the preceding context. As summarized in Table 1, the ranks of individual tokens remain largely stable, with fewer than 10% of tokens shifting rank in most cases. RL therefore selectively modifies the probabilities of a sparse set of critical tokens while leaving the bulk of the reasoning process untouched.

The three steps converge on a single, sharply stated conclusion—our first contribution: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains relatively stable during training.*

Beyond deepening our understanding of RL training dynamics, this conclusion directly supplies the ingredients of our theoretical framework: the stability of per-pattern success rates becomes a modeling assumption, and the “pattern-distribution reshaping” view becomes the object of our dynamical analysis.

Table 1: Fraction of tokens whose probability rank shifts after RL. Across tasks and training steps, only a small fraction (well under 10%) of tokens change rank, evidence that RL edits a sparse set of critical tokens.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

5 Theoretical Characterization of RL Training Dynamics

This section delivers our second contribution: a theoretical framework that explains the empirical findings above. We first abstract the model’s reasoning procedure as a two-step process—selecting a reasoning pattern, then deducing an answer—in which, consistent with our experiments, RL specializes in optimizing the first step. We then use this framework to explain RL under the two reward types. **For RLVR**, we prove convergence to the highest-success-rate pattern and expose two regimes with sharply different optimization behavior (Theorems 5.3 and 5.4), formalizing the empirical observation that RLVR’s gains depend on base-model quality. **For RLIF**, we explain both why internal rewards can raise performance and why they may ultimately drive the model below its starting point (Proposition 5.2 and Theorem 5.6), matching the observed instability. Each result is stated so that its correspondence to a specific empirical finding is explicit.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We formalize the reasoning process as follows: given a question q , the model (1) selects a reasoning pattern r from candidate patterns $\mathcal{R} = \{r_1, r_2, \dots\}$ and (2) generates a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ accordingly. The model selects reasoning pattern r_i with probability $p(r_i|q)$, and each reasoning pattern has a distinct success rate. Within this framework, we can reformulate the RLIF optimization objective (Eq. 3.3) as follows¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

Policy Parameterization. Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ represent the set of output sequences. Given a question q , the language model produces a distribution over output sequences $y \in \mathcal{Y}$ autoregressively:

$$\text{(general policy)} \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is a function parameterized by θ , and the model predicts the l -th token based on the previous context $y_{<l}$. The next token follows the distribution $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ for $z = f_\theta(x, y_{<l})$.

Due to the complexity of practical models, establishing optimization guarantees for understanding RL training dynamics has proven very challenging [3, 23]. Following previous works [27, 24, 10, 45], we consider a *tabular policy* parameterization, which can be viewed as a special case of Eq. 5.2 where each output is assigned its own trainable logit for the corresponding last token, i.e., for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$\text{(tabular policy)} \quad \pi_\theta(y_l|x, y_{<l}) = \pi_\theta(y_l|y_{l-1}) = \text{softmax}(\theta_{:,y_{l-1}})_{y_l}, \quad (5.3)$$

where $\theta_{:,y_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to y_{l-1} .

Optimization for RL. For our analysis of training dynamics, we consider the tabular policy from Eq. 5.3 with the optimization objective $\phi_{\text{RL}}(\theta)$ in Eq. 3.1. We analyze the policy gradient in the small learning rate limit using gradient flow:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

where $\theta(t)$ represents the parameters of the policy $\pi_{\theta(t)}$ at training time t , initialized with $\theta(0) = \theta_{\text{ref}}$.

Recall the empirical finding that per-pattern success rates stay stable during training; we now turn that finding into the modeling assumption our theory rests on. We attribute the stability to architectural constraints: optimizing the mapping from questions to reasoning patterns is substantially easier than optimizing the path from reasoning patterns to final answers. This motivates the following assumption:

Assumption 5.1. *The success rate for each reasoning pattern $r_i \in \mathcal{R}$ to provide the correct answer r^* , defined as $\pi_\theta(a^*|q, r_i)$, remains constant during training.*

In the remaining part of this section, we adopt Assumption 5.1 and use $p^*(r) = \pi_\theta(a^*|q, r)$ to denote the fixed success rate for the given pattern r . We first derive the optimal policy for the RL objective (Eq. 3.1) under the general autoregressive policy (Eq. 5.2). We then analyze RLVR and RLIF training dynamics using the tabular policy (Eq. 5.3).

5.2 The Optimal Policy for RL

Our empirical results show that RLVR steadily improves performance by incentivizing higher-success-rate patterns, whereas RLIF improves only unstably. As the first theoretical step, we characterize the *optimal policy* for both objectives, which already reveals why the two rewards behave so differently:

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

Proposition 5.2. Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the reasoning path reward. Specifically:

- For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution, i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

Proposition 5.2 already explains the core empirical contrast. Assuming π_{opt} lies within the general autoregressive policy space (which always holds for both LLMs and the tabular policies discussed later), the RLVR-optimized model increases the probability of a reasoning pattern $\mathbf{r}$ precisely when its success rate $p^*(\mathbf{r})$ is high enough that $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$. For RLIF, by contrast, $R_{\text{RLIF}}(\mathbf{r})$ depends only on the confidence of the final-answer distribution and cannot distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. The consequence is exactly the divergence we observed: the RLVR optimum consistently favors accuracy, whereas the RLIF optimum offers no such guarantee over the base model.

Moreover, since β is typically small (e.g., 0.001) in practice [26, 5, 44], as $\beta \rightarrow 0$, we have:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

In this limit, the policy converges to a deterministic strategy that always selects the reasoning pattern $\mathbf{r}$ with the highest $R(\mathbf{r})$, regardless of how the initialized reference model chooses patterns. The optimum alone, however, does not tell the full story: because of the non-convex landscape, it leaves open *how* RLVR reliably reaches high-reward patterns and *why* RLIF can improve early yet fail later. We answer both questions next by analyzing the training *dynamics* of the two methods under a tabular policy (Eq. 5.3)—the technical heart of this contribution.

5.3 Training Dynamic Analysis for RLVR

Our experiments showed that RLVR’s improvements depend heavily on the capacity of the base model; this subsection makes that dependence precise. We track the probability of the *optimal reasoning pattern*—the pattern $\mathbf{r}^*$ with the highest success rate for reaching the correct answer $\mathbf{a}^*$, $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$ —and show, via gradient flow, that RLVR training falls into one of two regimes. In Regime 1, $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases steadily until it converges to 1. In Regime 2, the model first passes through an *entanglement stage* in which a suboptimal pattern $\mathbf{r}' \neq \mathbf{r}^*$ slows optimization of $\mathbf{r}^*$; only after this stage does it transition into Regime 1 and finally converge to $\mathbf{r}^*$. The two theorems below correspond to the “strong base model” and “weak base model” cases, respectively.

Theorem 5.3 (Regime 1: Sufficient Condition for Efficient Convergence). Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$, defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$, satisfies:

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

Theorem 5.3 captures the favorable case: the base model is strong enough that its overall accuracy already exceeds the success rate of every non-optimal pattern (Eq 5.7). RLVR then guides the model to the optimal pattern at rate $\mathcal{O}(1/\epsilon)$, reaching an overall accuracy close to the success rate of $\mathbf{r}^*$. This is the theoretical counterpart of the smooth, monotone RLVR curves observed empirically.

Practice, however, shows that RLVR optimization is sometimes hard [44, 36], typically when the base model is weaker. Theorem 5.4 formalizes this second regime, in which the *suboptimal pattern with the second-highest success rate*, $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, entangles with $\mathbf{r}^*$ and slows its optimization:

Theorem 5.4 (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(Regime\ 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})} \quad (5.9)$$

with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.

For tractability, Theorem 5.4 treats the special regime in which only the optimal and suboptimal patterns have success rates above the overall accuracy; we expect the analysis to extend, with more effort, to any setting where *at least* two patterns beat the average.

The key takeaway of Theorem 5.4 is a sharp separation of timescales. RLVR needs T_0 steps merely to push the overall success rate past that of the suboptimal pattern—i.e., to *enter* the favorable regime of Theorem 5.3. Whereas Theorem 5.3’s convergence time T_1 is polynomial in $1/\epsilon$, the entry time T_0 can grow super-exponentially in $\gamma_{\pi_{\text{ref}}}$, the ratio between the total success rate through $\mathcal{R}/\{\mathbf{r}'\}$ and that through $\mathbf{r}^*$. When the base model assigns the optimal pattern a very small selection probability, $\gamma_{\pi_{\text{ref}}}$ is large and T_0 is prohibitive—explaining, in our framework, why weak base models can appear stuck for a long time before RLVR takes off. We call this the entanglement stage.

Both regimes are illustrated with case studies and simulations in Section 6, our third contribution.

5.4 Theoretical Explanation for RLIF

We now complete the RLIF side of the framework. Proposition 5.2 already showed that RLIF, depending only on the confidence of the final-answer distribution, cannot tell $\mathbf{a}^*$ from incorrect alternatives, hinting that RLIF may eventually degrade below the base model—consistent with our empirical observations. This raises an apparent paradox, since prior work reports that RLIF *does* improve models without external rewards Agarwal et al. [4], Zhao et al. [47]. We resolve the paradox here by explaining why RLIF helps a well-trained LLM at the *start* of training even though it can hurt later.

First, we consider a well-trained LLM satisfying the following assumption:

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$.*

Since majority voting [35] has proven to be an effective method for improving model performance, a well-trained LLMs typically assign a higher probability to the correct answer $\mathbf{a}^*$ than to incorrect answers, Assumption 5.5 is likely to hold for modern LLMs. Under this constraint, we consider the case that $|\mathcal{A}| = 2$, the success rate for each reasoning path follows a uniform distribution and the reasoning pattern selection is high-entropy². In this case, we analyze the overall accuracy dynamics at $t = 0$, yielding the following theorem:

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.
2. With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

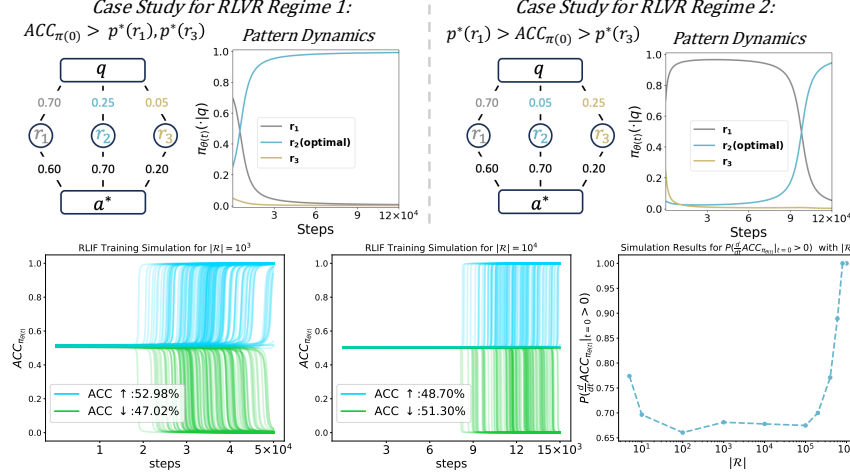


Figure 2: Validation of the theory (our third contribution): RLVR case studies (upper) and RLIF training simulations (bottom). Upper: the two predicted RLVR regimes—rapid convergence (left) and the entanglement phase (right). Bottom: for RLIF across varying $|\mathcal{R}|$, the probabilities of converging to different states (left) and the probability of an initial performance gain (right). Each panel matches a specific theorem, as detailed in the text.

Theorem 5.6 resolves the paradox in two parts, and together they mirror the rise-then-fall RLIF curves observed in our experiments. The first part shows that overall accuracy *increases* at initial-ization, giving a theoretical account of RLIF’s early gains. The second part exposes the danger that follows: since Proposition 5.2 implies that with $\beta = 0$ the policy converges deterministically to the pattern maximizing $R_{\text{RLIF}}(\mathbf{r})$, and this maximizer coincides with the *minimizer* of the success rate ($\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$) with probability 0.5, there is a 50% chance the model ultimately locks onto the least accurate reasoning path. RLIF thus helps at first (positive accuracy derivative) but, with continued training, can converge to low-accuracy patterns and end up worse than the base model—exactly the failure mode our experiments display.

6 Case Studies and Numerical Simulations

Our third contribution closes the loop: we validate every prediction of the framework with targeted case studies and simulations. For RLVR, we reproduce the two training regimes of Theorems 5.3 and 5.4 (Fig. 2, upper). For RLIF, we run multiple training simulations in which each reasoning path’s success rate is drawn from a uniform distribution, confirming our analysis of both RLIF’s initial-step behavior and its convergence (Fig. 2, bottom). We take each in turn.

6.1 Case Studies for RLVR

Recall the two RLVR regimes of Section 5.3: in Regime 1 (Eq 5.7) the probability of the optimal pattern $\pi_{\theta}(\mathbf{r}^*|q)$ increases monotonically until convergence, while in Regime 2 (Eq 5.8) an initial entanglement phase delays convergence to $\mathbf{r}^*$ for a substantially longer period. The two case studies below show each regime arising exactly as predicted:

- As demonstrated in Fig. 2 (upper left), when the reference model satisfies the conditions in Eq 5.7, where the overall accuracy exceeds the success rates of all non-optimal reasoning patterns, the model converges rapidly to the optimal reasoning pattern.
- In the second case, corresponding to Theorem 5.4, when a suboptimal reasoning pattern $\mathbf{r}'$ achieves a higher success rate than the initial overall accuracy, the model first experiences an entanglement phase. A transition period T_0 (defined in Theorem 5.4) must elapse before the model enters the rapid optimization phase characteristic of case 1. Fig. 2 (upper right) illustrates a scenario where $\gamma_{\pi_{\text{ref}}}$ (defined in Eq 5.9) is large ($\gamma_{\pi_{\text{ref}}} = 6$, as $(0.25 + 0.05)/0.05 = 6$). In such cases, the transition time T_0 , which scales with $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$, becomes prohibitively long, significantly delaying convergence to the optimal reasoning pattern. The pattern dynamics in Fig. 2 clearly demonstrate such entanglement stage, align well with our theoretical result.

6.2 Numerical Simulations For RLIF

In Section 5.4, Theorem 5.6 made two predictions for RLIF (under Assumption 5.5 and the base-model distribution specified there): as $|\mathcal{R}| \rightarrow \infty$, the accuracy derivative at $t = 0$ is positive with probability 1, and there is a 50% probability of eventual convergence to the least accurate reasoning path. Our simulations confirm both. Fig. 2 (bottom left) sweeps $|\mathcal{R}|$ and shows the probability of converging to an improved state ($ACC \uparrow$) is approximately equal to that of a degraded state ($ACC \downarrow$), matching the second prediction; Fig. 2 (bottom right) shows that for large $|\mathcal{R}|$ the probability of an initial performance gain approaches 1, matching the first. The framework’s account of RLIF is therefore validated end to end.

7 Conclusions and Limitations

This work explains RL training dynamics in LLMs through the unifying lens of reasoning pattern selection, and its value rests on three contributions. *Empirically*, our pattern-level and token-level analysis establishes that RL optimizes a sparse subset of critical tokens and reshapes the distribution over reasoning patterns, while per-pattern success rates stay stable. *Theoretically*, our two-stage framework turns this observation into a formal account of RLVR (two convergence regimes governed by base-model quality) and RLIF (early gains followed by possible degradation). *Empirically again*, case studies and simulations confirm each prediction. Together they close the loop from observation to theory to re-validation and yield concrete guidance for RL-based post-training. Three limitations mark the path forward: the interpretability of LLM-identified reasoning patterns needs further validation, the framework could be extended to more complex real-world reasoning, and the analysis could be generalized beyond our specific base-model assumptions.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

The main-paper pattern-dynamics analysis (Section 4.2) used Number Theory tasks from MATH; here we replicate it on Geometry tasks (Figure 3) to show the finding is not task-specific. The same picture holds: under RLVR the model gradually adopts higher-accuracy patterns, whereas RLIF’s pattern selection remains unstable. This directly reinforces our first contribution and is consistent with both the empirical and theoretical results in the main paper.

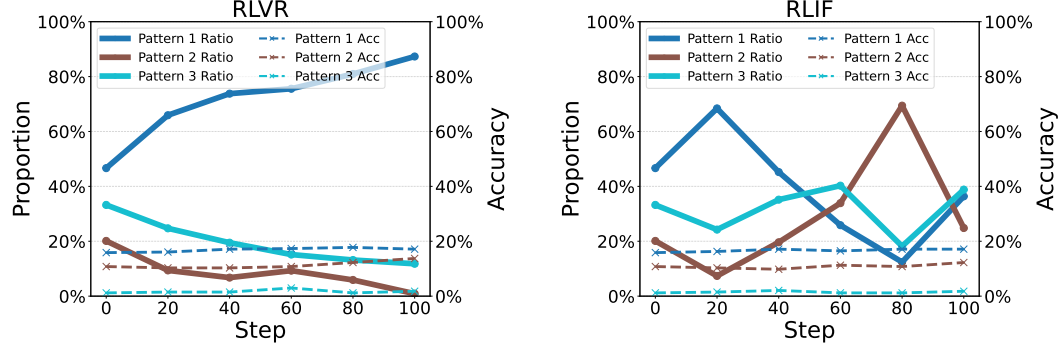


Figure 3: Pattern-distribution dynamics and per-pattern accuracy on Geometry tasks, replicating the main-paper finding on a second task type. RLVR concentrates on the highest-accuracy pattern (Pattern 1), while RLIF stays unstable.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To test whether our findings generalize across models, we run supplementary experiments that stress-test both the experimental insights and the theoretical conclusions (Figure 4). We use Qwen-2.5-7B-Instruct [39] as the base model and compare it with Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source variant enhanced with RLVR for mathematical reasoning. The evaluation spans diverse mathematical domains—number theory, geometry, algebra, calculus, counting and probability—using challenging problems from the MATH dataset [16], and further includes complex problems from AMC23 [1]. Every domain tells the same story that grounds our first contribution: RLVR raises the frequency of high-accuracy reasoning patterns and suppresses less effective ones. Task-specific details are given in Appendix B.3.

Across all these tasks, the highest-success-rate patterns consistently become more prevalent after RLVR, reinforcing our first contribution (Section 4) and its formalization (Section 5). Notably, in tasks such as Algebra and AMC23-19 (Figure 4), per-pattern success rates stay consistent before and after RLVR—which both aligns with our earlier observations and supplies direct empirical support for Assumption 5.1, the linchpin of our theoretical analysis.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To probe whether our theory holds at scale, we further experiment on a larger model, QwQ-32B [32], which is built on Qwen-2.5-32B-Instruct [39] and greatly strengthened via RLVR. We measure the reasoning-pattern shift between the two models on four tasks: number theory and geometry from MATH [16], a task from AIME 2024 [7], and a task from OlympiadBench [15].

The result extends our contributions to the 32B scale: most of QwQ-32B’s reasoning patterns coincide with the most accurate patterns of Qwen-2.5-32B, exactly as our theory predicts. We omit QwQ’s per-pattern accuracy because API-only access prevents us from extracting CoT data and letting the base model continue generation, so pattern-level (as opposed to model-level) accuracy cannot be obtained; see “Accuracy Analysis” in Appendix B.2.

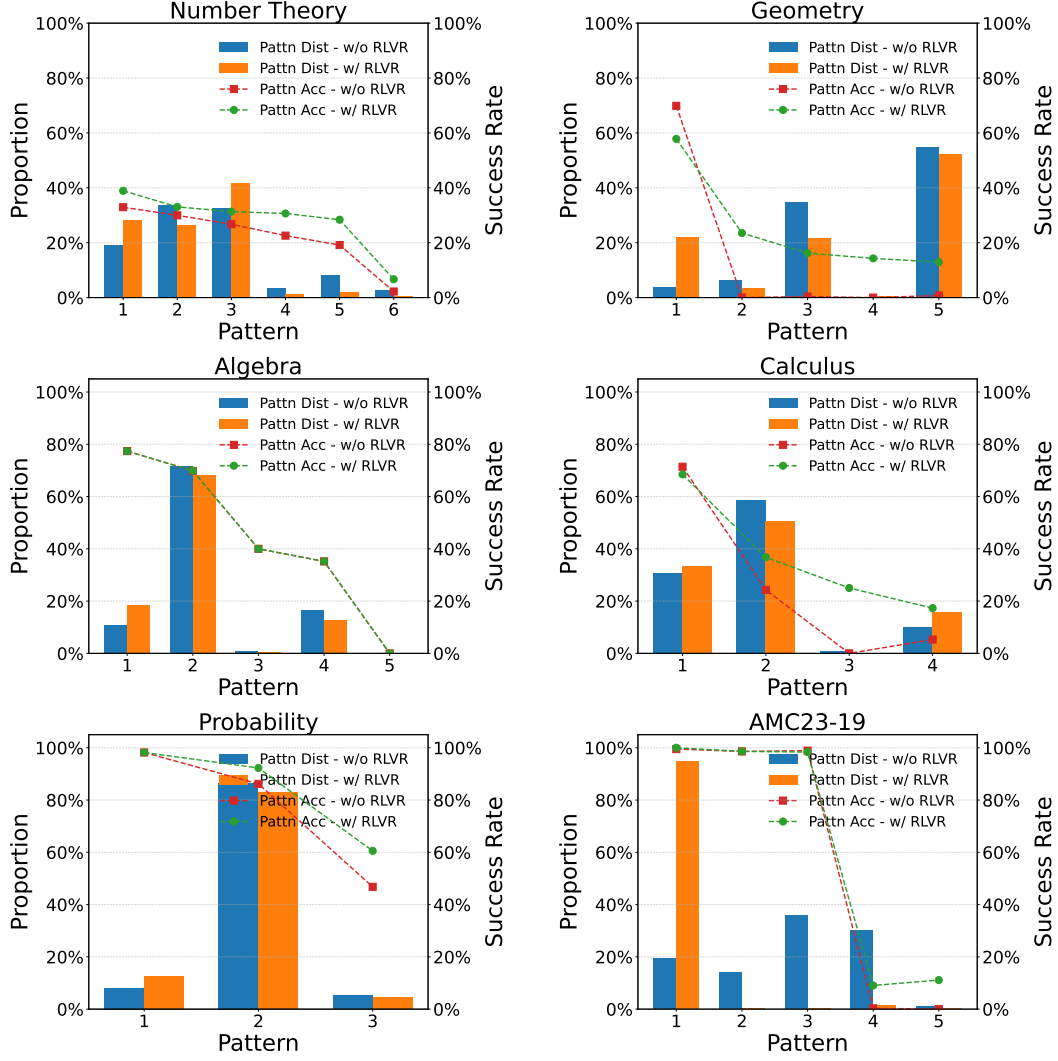
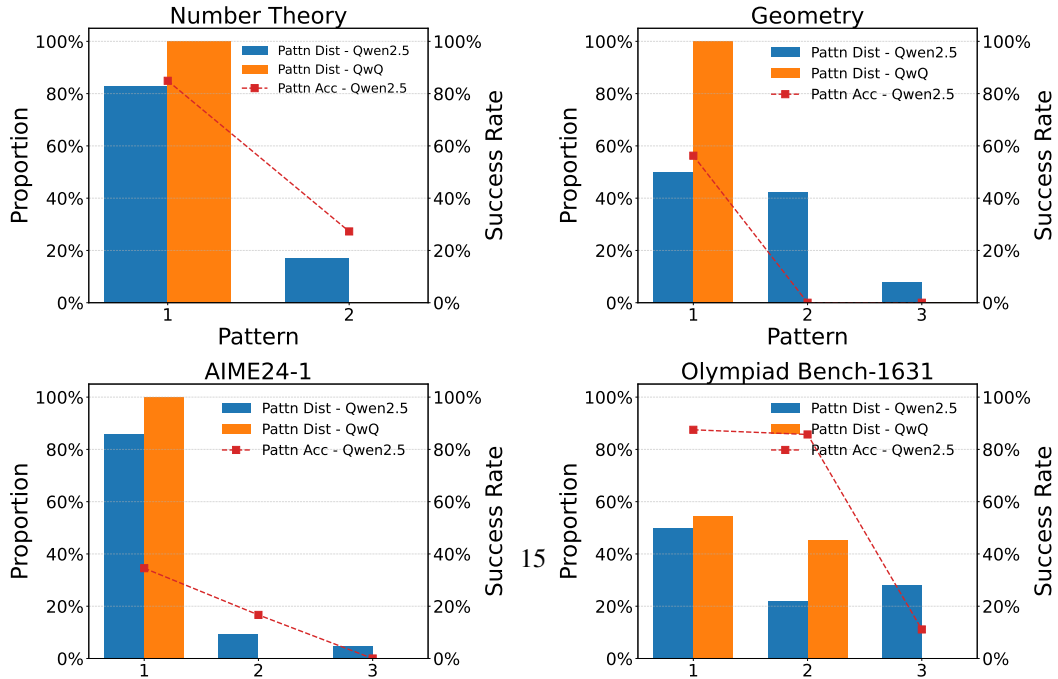


Figure 4: Reasoning-pattern distribution and per-pattern success rate with/without RLVR across additional tasks, all consistent with our experimental insights and theoretical conclusions. Bars (Pattern Dist) give each pattern’s share of all patterns; dots (Pattern Acc) give its accuracy/success rate.



570 B Experimental Details

571 B.1 Experimental Setup

572 This appendix documents the setup behind each experiment reported in the main paper, so that the
573 evidence for our three contributions is fully reproducible.

574 **Entire Training Procedures:** We select Qwen2.5-3B as the base model and train it on the MATH
575 dataset with both RLVR and RLIF. We use the verl [31] framework and keep all parameter settings
576 identical to the MATH example it provides; for RLIF we modify only the reward computation. Setting
577 different random seeds yields three training curves per method.

578 **Token-level Analysis:** This experiment underpins the sparse-token claim of our first contribution.
579 We use Qwen2.5-3B and the RL-enhanced models above. We randomly sample 400 test questions
580 each from GSM8K and MATH, and use the full AIME24 dataset. For each question we first sample
581 an answer from the base model, then concatenate that answer to the question and feed it to the
582 RL-enhanced model as a new prompt. Using the OpenAI library interface, we compute each token’s
583 probability and its rank among all possible tokens at that position; comparing ranks before and after
584 RL yields the proportion of tokens whose rank changed.

585 **Reasoning-pattern Level Analysis:** This experiment supports the pattern-resaping claim of our
586 first contribution and runs in two modes. First, we study the pattern-distribution *dynamics* during
587 training using our own trained models, Qwen2.5-3B and its RL-enhanced versions, evaluating every
588 20 steps. Second, to test whether the conclusions transfer to other and larger base models, we compare
589 the open-source Qwen-2.5-7B-SimpleRL-Zoo and QwQ-32B against their base models; for these we
590 only have access to outputs before and after RL, so we examine the before/after change in pattern
591 distribution rather than its full training trajectory.

592 B.2 Reasoning Patterns Analysis Pipeline

593 **Model Inference:** We randomly sample questions of different task types (Geometry, Algebra, etc.)
594 and difficulty levels (Level 4, Level 5) from datasets including MATH. Per question, the model
595 samples 1024 answers in the pattern-dynamics task, and between 64 and 1024 answers (depending on
596 task category) in the before/after task. These responses feed the pattern extraction and classification
597 steps below.

598 **Reasoning Patterns Extraction:** We sample a subset of base-model responses and use the GPT-4o
599 [19] API to summarize pattern categories from them; the full extraction prompt is in Appendix D.1.
600 To make extraction deterministic, we set GPT-4o’s temperature to 0.

601 **Responses Classification:** We again use GPT-4o (temperature 0) to classify every response into
602 one of the extracted pattern categories; the full classification prompt is in Appendix D.2.

603 **Accuracy Analysis:** To isolate the effect of the pattern *distribution* on accuracy—the crux of our
604 first contribution—we hold the reasoning process fixed and vary only which pattern is followed.
605 Concretely, we build new prompts by concatenating the original prompt with a partial base-model
606 response that represents a specific reasoning pattern: we drop the sentence containing the final answer,
607 append the preceding text to the question, and feed this to the RL-enhanced model to continue.
608 Because the reasoning is fixed in advance, any change in the resulting answer accuracy is attributable
609 to the pattern rather than to a changed reasoning process; we report this accuracy as the pattern’s
610 success rate.

611 B.3 Details for Each Task

612 For completeness, we detail the procedure for each task below, including task descriptions and
613 example reasoning patterns (Table 2); example prompts are in Appendix D.3.

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Examples of the reasoning patterns extracted for each task, illustrating the pattern vocabulary that our analysis tracks.

Number Theory: This task concerns coins distributed across bags that initially hold equal amounts; after receiving extra coins and redistributing them equally, the total must exceed a specified value while remaining evenly divisible, and the model must find the minimum initial count per bag. We derive it from a level-5 MATH problem (id: test/number_theory/1055) [16] and build a 32-variant evaluation set by varying parameters such as bag count and coin totals. We use one-shot prompting with a simple, unrelated example to fix the “boxed{ }” answer format without steering the reasoning approach. For evaluation, we take 20 questions with 4 responses each for pattern extraction and sample 64 responses per question, then classify all responses by the extracted patterns to compare the pattern distribution before and after RL.

Geometry: In the geometry task, the model is given two side lengths of a scalene triangle and asked how many integer centimeter values the third side can take. It derives from a level-4 MATH geometry problem (id: test/geometry/1046). Evaluation mirrors the Number Theory pipeline: a 32-question synthetic set, 20 questions with 4 base-model (Qwen2.5-7b-Instruct) responses each for pattern extraction, and then a with/without-RLVR comparison over all evaluation questions with 64 responses each.

Algebra: The algebra task is a can-recycling problem: given a stock of old cans and the rule that n old cans yield one new can, the model must compute how many cans can ultimately be produced. Its difficulty lies in the leftover cans that cannot be evenly divided at each step and may later be pooled for further recycling. The template is a level-4 MATH algebra problem (id: test/algebra/2768). This

and all subsequent supplementary tasks use the same settings as the tasks above (dataset size, number of samples, etc.).

Calculus: The calculus task is deliberately simple: we present a single-variable trigonometric equation and its domain and ask for the number of solutions. Though easy, it probes the model’s basic calculus competence. Its prototype is a level-4 MATH pre-calculus problem (id: test/precalculus/1140).

Counting and Probability: We also study a counting-and-probability task. Given white and black balls, two are drawn at random, and we specify the probability that one is black and the other white together with the count of one color; the model must find the minimum count of the other color. It derives from a level-4 MATH counting-and-probability problem (id: test/counting_and_probability/79).

AMC23 Question: We use question 19 of AMC23 [1], sampling 48 responses for pattern extraction and 1024 responses for pattern analysis.

B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

We run four tasks. Because API-only testing constrains data access, we select one question per task, sample 64 responses from each of the two models, and perform pattern extraction and classification directly on these answers. From MATH we use question 1055 (number theory) and question 1046 (geometry); the remaining two are problem 1 from AIME24 and question 1631 from OlympiadBench. Example patterns appear in Table 2.

C Deferred Proofs

This appendix supplies the full proofs behind our second (theoretical) contribution, in the order they appear in the main text: Proposition 5.2 (optimal policy), Theorem 5.3 (RLVR fast-convergence regime), Theorem 5.4 (RLVR entanglement regime), and Theorem 5.6 (RLIF early gain and possible collapse).

C.1 Proof of Proposition 5.2

Proof of Proposition 5.2. In this proof, we utilize the proof techniques in [26], recall that the optimization objective of RL is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

Under our framework, as state in section 5.1, we consider a policy conduct reasoning by first sample $\mathbf{r}_i \in \mathcal{R}$ based on $\pi_{\theta}(\mathbf{r}_i|\mathbf{q})$ and then provide the final answer $\mathbf{a} \in \mathcal{A}$ by $\pi_{\theta}(\mathbf{a}|\mathbf{r}_i)$, the reward $\phi_{\text{RL}}(\theta)$ can be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$, then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot|\mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[\ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

Where the third equation is by Assumption 5.1 and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the partition constant that ensures

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta}(\mathbf{r}|\mathbf{q}),$$

664 is a valid probability distribution such that $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$. Since Z is not a function of $\mathbf{r}$.

665 Therefore, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned} & \max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ & = \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

666 By the properties of KL-divergence, we know that the optimal policy for the KL-constrained reward
667 maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

668 specifically, for RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$, for RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \log |\mathcal{A}| \right)}_c, \end{aligned}$$

669 where the second term is the same for all $\mathbf{r} \in |\mathcal{R}|$, for RLIF, the optimal policy can be written as

$$\begin{aligned} \pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i)) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp \left(\frac{1}{\beta} (R'(\mathbf{r}) - c) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \end{aligned}$$

670 So we can write $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r}))$. This concludes the proof
671 of the theorem. $\square$

672 C.2 Dynamics of Tabular Policy

673 Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the
674 gradient dynamics of θ can be computed as

$$\begin{aligned} \frac{d}{dt} \theta(t) &= \nabla \phi_{\text{RL}}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})], \end{aligned}$$

675 Where $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to $\mathbf{q}$, and
676 $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned} \frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i} \right] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})) \right] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \end{aligned}$$

677 where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we
 678 have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t}$$

679 C.3 Proof of Theorem 5.3

680 To proof Theorem 5.3, here we first prove that $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4), then we
 681 establish a lower bound for $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$ (Eq C.5), finally we derive the final bound for t stated in
 682 Theorem 5.3.

683 *Proof of Theorem 5.3.* Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

684 For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}), \end{aligned}$$

685 and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) p^*(\mathbf{r}_i) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \quad (\text{C.4}) \end{aligned}$$

686 As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern
 687 $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$
 688 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \geq 0. \quad (\text{C.5}) \end{aligned}$$

689 As $\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})$ holds for $t \geq 0$, define $C =$
 690 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^*|\mathbf{q})$:

$$\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.$$

691 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})}}.$$

692 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \epsilon.$$

693 This concludes the proof of the theorem. $\square$

694 C.4 Proof of Theorem 5.4

695 To proof Theorem 5.4, we first establish an upper bound for $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ (Eq C.6), then we analyze
 696 the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$ (Eq C.7) and prove that for $t > T_0$, $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$.

697 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
 698 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned} \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}')) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\ &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2. \end{aligned}$$

699 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

700 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned} \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\ &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0 \end{aligned}$$

701 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 702 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

703 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned} \frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}} \end{aligned}$$

704 Then we have:

$$\frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

705 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

706 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

707 Let $C_2 = 1/\Delta$, $C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

708 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

709 Finally, for $t > T_0$ we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

710 This concludes the proof of the theorem. $\square$

711 **C.5 Proof of Theorem 5.6**

712 *Proof of Theorem 5.6.* First, as we consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$, Assumption
 713 5.5 holds means $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of overall accuracy can be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|q) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|q) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, q)) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

714 where $\text{Con}_{\theta(t)} = - \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i|q) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, q))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
 715 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i|q) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, q)) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_j, q)) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, q)) \right). \end{aligned}$$

716 As we consider the case $|\mathcal{R}| \rightarrow +\infty$ and the success rate $p^*(\mathbf{r})$ are i.i.d. samples from $U[0, 1]$.
 717 By the Central limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r}))$ is concentrated
 718 around its expectation 0.5, the deviation is of order $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. We define this deviation as
 719 $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ such that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$.

720 According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$,
 721 the empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$
 722 converges to the density: $f(x) = 1 + 12\delta(x - 0.5), x \in [0, 1]$. As $|\mathcal{R}| \rightarrow \infty$, we can approximate
 723 the summation over $\mathbf{r}$ with an integral over this limiting distribution $f(x)$. The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3} \delta > 0$$

724 Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this holds with probability 1, proving the
 725 first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size
 $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

726 The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$
 727 is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the
 one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

728 Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

729 We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting
 730 distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and
 731 maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables.
 732 Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- 733 • Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) =$
734 $1 - 6\delta$.
735 • Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) =$
736 $1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta\right) = \frac{1}{2}$$

737 This concludes the proof of the second statement.

738

□

739 D Full Prompts

740 D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

741

Example processes from input
 Common elements
 Pattern 2:
 Name and description
 Defining characteristics
 Example processes from input
 Common elements
 ...
 There are reasoning processes:
 {extracted_responses}
 Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

742

743 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.
 {patterns_categories}
 Here are the reasoning processes to classify:
 {extracted_responses}
 Please output your classification in this JSON format:

744

745 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system
 You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_endl>
 <lim_start>
 user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_endl>
 <lim_start>assistant
 Let me solve this step by step.
 <think>

746

Prompt Example for Long Arithmetic

<lim_start>system
 You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_endl>
 <lim_start>user
 Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_endl>
 <lim_start>assistant
 Let me solve this step by step.
 <think>

747

Prompt Example for Number Theory

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

748

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

749

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

750

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

751

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

752

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of ' $8^5 \cdot 5^{10} \cdot 15^5$ '?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

753

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a 9 -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

754

Prompt Example for OlympiadBench-Q1631

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

755